PRIMATES, 25(2): 204–217, April 1984

Mother-infant Separation and Reunion in the Great Apes

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ABSTRACT. One mother-infant dyad of each species of great ape, orang-utan, chimpanzee and gorilla, with an additional conspecific cagemate, were studied for the effects of maternal separation and reunion. High levels of agitation were observed in all infants immediately upon separation, followed by a longer period of behavioral depression with continuing, intermittent agitation. A compatible cagemate seemed to attenuate, to some degree, the behavioral depression reaction. Initial detachment following reunion with the mother occurred for all the great ape infants studied. Subsequent intensification of mother-infant attachment occurred during reunion for all three species. The findings of intermittent agitation during separation and the initial detachment upon reunion with the mother are not generally reported for monkeys, but are reported for human children. These data suggest that certain responses to maternal separation and reunion occur, to some degree, among all primates that have been studied, whereas other responses seen among apes and humans are not generally reported for monkeys.

INTRODUCTION

A dominant characteristic of the Order Primates is that the mother-infant dyad forms a strong and enduring bond. Extensive research has been conducted on maternal separation and reunion in monkeys with emphasis on the infant's response to separation from its mother (MCKENNA, 1979; MINEKA & SUOMI, 1978; REITE, 1977). SPITZ (1946) originally conducted studies on early human attachment and the debilitating effects on children when separated from the mother figure. Two distinct phases, consisting of a period of agitation and crying followed by a period of withdrawal and depression, were reported for children in response to maternal separation. ROBERTSON and BOWLBY (1952) and BOWLBY (1960) described the reactions of children who had been separated from their mothers as typically occurring in three phases. A protest phase, characterized by distress, agitation and crying, occurred immediately upon separation. Following the protest phase, a phase of despair developed, characterized by reduced activity, social withdrawal and depressed posture. Upon reunion, a detachment phase occurred in which the infant showed rejection and hostility toward the mother. In the area of nonhuman primatology, laboratory studies on several species of monkeys frequently have identified a biphasic response to maternal separation, consisting of agitated movement followed by behavioral depression (HINDE, SPENCER-BOOTH & BRUCE, 1966; JENSEN & TOLMAN, 1962; ROSENBLUM & KAUFMAN, 1968; SEAY, HANSEN & HARLOW, 1962; SEAY & HARLOW, 1965).

Due to similar responses to maternal separation between monkeys and humans, several authors have proposed that monkeys could serve as useful models for the systematic study of human separation and depression (MCKINNEY, SUOMI & HARLOW, 1973). However, others have challenged this proposal because of a certain amount of variability in the response of monkeys to separation from mothers. One major difference between the responses of mon-

keys and children is the absence of a detachment reaction upon reunion with the mother when the monkey infant is in a free-access situation (MINEKA & SUOMI, 1978). Failure to observe detachment in most infant monkeys that have been studied seems to be related to the conditions of testing (PLIMPTON & ROSENBLUM, in press).

Because of their somewhat closer biological relationship to humans, the great apes could provide additional information relevant to the interpretation of the behavioral phenomena found in humans and monkeys in response to maternal separation. Very little research of this type has been conducted with the great apes. However, the limited data that are available suggest that in these species, typical patterns of agitation followed by behavioral depression and apparent evidence of brief detachment frequently occur in response to maternal separation and reunion (HOFF et al., 1979; NADLER & CODNER, 1983; NADLER & GREEN, 1975). Although these reactions to maternal separation have been frequently observed, they may vary, depending on species, age, developmental stage and environmental conditions (LEWIS et al., 1976; MCKENNA, 1979; MITCHELL, MAPLE & ERWIN, 1979).

The purpose of the present study was to investigate the effects of maternal separation and reunion on a single subject of each great ape species, orang-utan, chimpanzee and gorilla. Since only one member of each species was studied, it is not possible to differentiate individual from species differences. Therefore, for the purpose of this study, species variability is disregarded and commonalities among the apes are examined for their relationship to the earlier data presented on monkeys and humans. Moreover, there was a third conspecific living with each mother-infant dyad prior to and during the current study. The purpose of the third animal was to provide the infant with a familiar cagemate in order, perhaps, to attenuate the stress of separation and to serve as a potential surrogate care-giver for the infant during the transfer of attachment and the subsequent study of detachment.

METHODS

SUBJECTS

The subjects were three captive-born great ape infants and their wild-born, multiparous mothers which were observed throughout the infants' first year of life as part of an ongoing investigation of mother-infant relations and infant development in the great apes (HOFF, NADLER & MAPLE, 1981; MILLER & NADLER, 1981). All mothers had lived in captivity at the Yerkes Regional Primate Research Center of Emory University for at least 15 years and previously had reared offspring successfully. The female orang-utan (*Pongo pygmaeus abelii*) infant was 14 months of age at the start of the study, and an unrelated, captive-born juvenile male, 4 years of age, had lived with this dyad for 14 months. The female chimpanzee (*Pan troglodytes*) infant was 13 months of age at the start of the study. An unrelated, wild-born adult female chimpanzee had lived with the mother-infant dyad for 13 months. The female gorilla (*Gorilla gorilla gorilla*) infant was 30 months of age at the start of the study; a captive-born, adult female sibling, 9 years of age, had lived with the dyad for 30 months.

PROCEDURE

The mother-infant dyads and cagemates were housed together in an indoor compartment and an outdoor enclosure, interconnected by a guillotine door [for a more detailed description

of the housing conditions refer to NADLER & CODNER (1983)]. In the present study, observations were made while the subjects were confined to the inner compartment. Testing procedures involved recording individual and social behaviors during (1) a two-week pre-separation period; (2) a two-week period of separation from the mother; and (3) a two-week period of reunion. The subjects were observed for 30 min, three to five times per week, throughout the six weeks. Behavioral data were collected on a modified HANSEN (1966) checklist by a trained observer who sat 5–10 ft in front of the subjects' cage. Behaviors of interest were scored as present or absent for each 1-min interval of the 30-interval (min) test. Although a variety of behavioral categories were scored, the categories considered in this paper are ventro-ventral contact, nipple contact, positive interactive behaviors with the cagemate, agitation and behavioral depression. Agitation in the infant was defined by distress vocalizations and active locomotor behaviors (i.e., active climbing for the orang-utan; active climbing and walking for the chimpanzee and gorilla). Behavioral depression was defined by a lack of locomotion (i.e., inactive hanging for the orang-utan; inactive sitting and reclining for the chimpanzee and gorilla). Detachment was defined by a failure of the infant to approach its mother upon reunion. Use of these terms is not intended to imply the presence of cognitive concomitants previously attributed to humans.

The physical separation of mother and infant was accomplished with a minimum of handling and trauma. In the case of the gorilla dyad, the mother was separated from the infant by closing the guillotine door when the two subjects were on opposite sides, and then removing the mother. The infant then was left in the cage in which it was born and reared, and which contained the adult female sibling. In both the chimpanzee and orang-utan dyads, the mother and infant were placed in a squeeze cage and the mother was anesthetized with ketamine hydrochloride (5 mg/kg). The infant was then taken from the mother and returned to the home cage, which contained the respective cagemate. In all subjects, observations were initiated immediately following the return of the infant to its cage. The mother was placed in a separate cage some distance from the infant. The infants and mothers were separated visually from each other and since the mothers did not vocalize, the separations can be considered as total. The chimpanzee and gorilla cagemates remained in the cage with the respective infants during the entire study. However, the juvenile male orang-utan was removed during testing only on the second week of the separation period. The juvenile male was removed for the duration of each observation session because of its persistent attempts to interact with the infant, which compromised our ability to evaluate independently the infant's activities related to separation. Prior to daily removal of the male, it was observed with the infant for several minutes for evidence of adoption. The possible effects of the juvenile male's absence during the second week's tests are considered below. Following the two-week period of separation, each mother was placed in a transport box and positioned in front of the infant's cage, providing the subjects with visual, but not physical, contact for a duration of 10 min. Following this initial assessment, the mother was released into the home cage with the infant and cagemate, and observations continued for 30 min. Observation sessions were continued for two weeks following the reunion.

RESULTS

The effects of a separation experience on the infant are assessed by behavioral changes in terms of the mother-infant relationship, cagemate-infant relationship, agitation and behav-

ioral depression. Ninety-five percent confidence intervals were established for behaviors related to the mother-infant relationship and cagemate-infant relationship prior to separation. Ninety-nine percent confidence intervals were established for behaviors related to agitation and behavioral depression. The more conservative test for significance was used for the latter types of behavior because of the near-zero levels and low variance observed during pre-separation. Since tests were not conducted daily, days in relation to critical events, e.g., separation and reunion, as well as test sessions are presented.

Mother-infant Relationship

The effects of a separation experience on the subsequent mother-infant relationship are indicated by changes in levels of mother-infant behaviors during the period of reunion in comparison to pre-separation levels. Increased levels of contact during reunion suggest an intensification of the mother-infant relationship.

Ventro-Ventral Contact and Nipple Contact

Pre-separation

At the time the present study was initiated, when the orang-utan, chimpanzee and gorilla infants were 14, 13 and 30 months of age, respectively, the mother-infant relationships appeared to be waning both in terms of ventro-ventral contact (Fig. 1) and nipple contact (Fig. 2) for all three species. During the two weeks prior to separation, the infants spent a considerable amount of time off the mothers, climbing on the cage and examining various objects in the cage. Ventro-ventral contact occurred relatively infrequently for the orang-utan (9.0%), chimpanzee (12.3%) and gorilla (5.7%) prior to separation. Nipple contact occurred even less frequently for all three species (5.7%, 5.0% and 3.3%, respectively) during the same two-week period.

Reunion

When the transport box containing the orang-utan mother was wheeled in front of the cage containing the juvenile male and infant, the male, with the infant clinging to it, came to the front of the cage and looked in the direction of the mother. The male then climbed to the top and rear of the cage, the farthest distance possible from the mother, and remained suspended therefrom. The infant glanced at the mother several times initially while maintaining its hold on the male, but made no effort to leave the male or approach the mother (detachment). The mother looked in the direction of the infant and shook the transport box in an apparent effort to gain closer proximity to the cage.

After 10 min had elapsed, the mother was released into the cage, looked toward the male and infant, and moved in their direction. The male remained pressed against the top of the cage, obviously avoiding the mother, and the infant continued to cling ventrally to the male (detachment). The mother did not attempt to retrieve the infant during this initial period of reunion, and the infant did not try to leave the male. This relationship continued for 10 min before the infant relinquished its hold on the male and approached and established ventral contact with its mother.

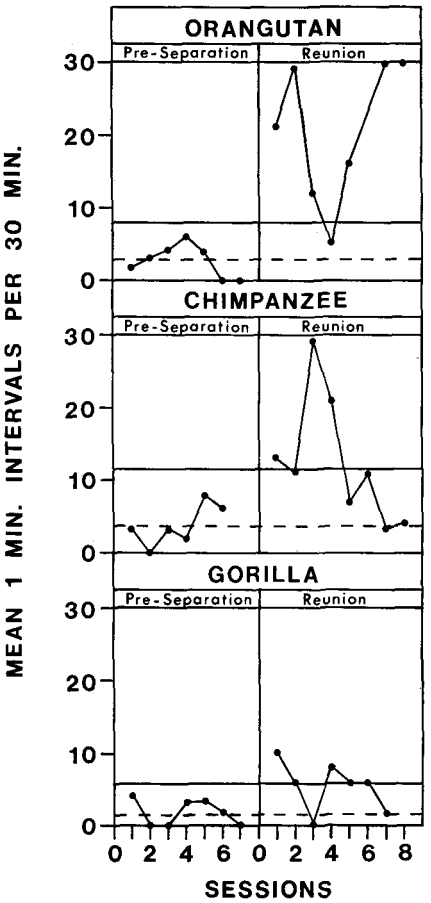


Fig. 1. Levels of ventro-ventral contact for orang-utan, chimpanzee and gorilla mother-infant dyads prior to separation and following reunion. Broken horizontal lines represent mean levels and solid horizontal lines represent 95% confidence levels for pre-separation levels.

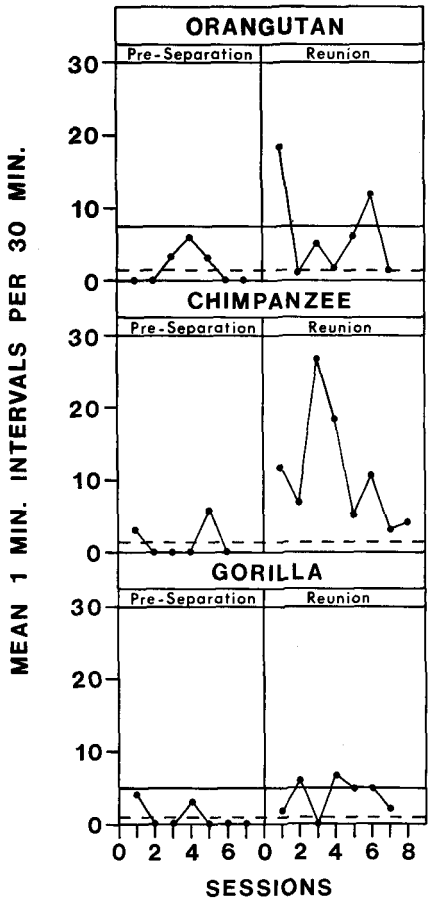


Fig. 2. Levels of nipple contact for orang-utan, chimpanzee and gorilla mother-infant dyads prior to separation and following reunion. Broken and solid horizontal lines are defined as in Figure 1.

Once mother-infant contact was reestablished, there was an intensification of mother-infant attachment, reflected by the measures of contact described above. Ventro-ventral contact between the orang-utan mother and infant increased significantly (S.D. = 2.2, $df = 6$, $p < .05$) for observation sessions 1–3, with a peak during session 2 (day 2) of reunion and a return to baseline by session 4 (day 7).

When the transport box containing the chimpanzee mother was placed in front of the cage containing the adult female and infant, the adult female cagemate first came to the front of the cage and looked in the direction of the mother. The mother looked in the direction of the infant and shook the transport box. When the infant looked in the direction of the mother, it walked bipedally toward the cagemate with outstretched arms while vocalizing in apparent distress. The infant then sat huddling in a collapsed posture and made no apparent effort to gain proximity with its mother (detachment). Immediately prior to the release of the mother

into the cage, the infant ran to the rear guillotine door and remained the farthest distance possible from the mother. When the mother was released into the cage, it immediately retrieved the infant, placing it in ventro-ventral contact. Ventro-ventral contact between the chimpanzee mother and infant increased significantly (S.D. $= 2.9$, $df = 5$, $p < .05$) for session 1 and sessions 3–4, with a peak on session 3 (day 3) and a return to baseline by session 5 (day 7).

When the transport box containing the gorilla mother was positioned in front of the cage containing the sibling female and infant, the mother began shaking the box furiously. The infant looked in the direction of the mother and immediately retreated by running to the rear of the cage (detachment). The infant slapped at the guillotine door in an apparent effort to open it. The infant then climbed to the top and rear of the cage while looking at the mother. The infant briefly initiated ventro-ventral contact with the cagemate. When the mother was released into the cage, it rushed to the rear of the cage. While the mother growled and looked in the direction of the infant, the infant avoided the mother by running to the opposite end of the cage (detachment). After 1 min had elapsed, however, the infant slowly approached the mother and initiated contact. After 15 min, the infant broke contact with and remained distant from the mother while holding itself. When the mother reached out to the infant, the infant slapped the mother and attempted to bite. Although ventro-ventral contact between the gorilla mother and infant increased significantly (S.D. $= 1.7$, $df = 6$, $p < .05$) for sessions 1–2 as compared to pre-separation levels, the relative increase was considerably less than that for the orang-utan and chimpanzee. While levels of ventro-ventral contact in the chimpanzee and gorilla remained near baseline throughout the second week of reunion, a second significant increase in ventro-ventral contact for the orang-utan occurred during sessions 5–7. (Although the orang-utan infant was alert and appeared to be healthy throughout the study, it was observed to be lethargic two days after the end of the study. It was taken from the mother, given a physical examination and placed in the Center's nursery for treatment of strongyloides hyperinfection. Despite the treatment, the infant died five days later of a massive pulmonary hemorrhage.)

Nipple contact for the orang-utan increased significantly (S.D. $= 2.4$, $df = 6$, $p < .05$) to a maximum level during session 1 (day 1) with a return to baseline by session 2 (day 2). Nipple contact for the chimpanzee increased significantly (S.D. $= 2.5$, $df = 5$, $p < .05$) during session 1 and sessions 3–4, with a peak during session 3 (day 3). While nipple contact showed a significant increase (S.D. $= 2.4$, $df = 6$, $p < .05$) for the gorilla during sessions 2 and 4, the levels were lower than those for the orang-utan and chimpanzee.

Cagemate-infant Relationship

The cagemate-infant relationship is observed for effects on the infant's reactions to maternal separation and subsequent reactions to reunion with its mother. The cagemate-infant relationship is assessed by observation of positive interactive behaviors as defined by social play, infant-initiated social examination and infant-initiated contact, including all infant-accepted ventro-ventral contact.

Pre-separation

During the two-week period prior to separation, relatively high levels (26.3%) of positive

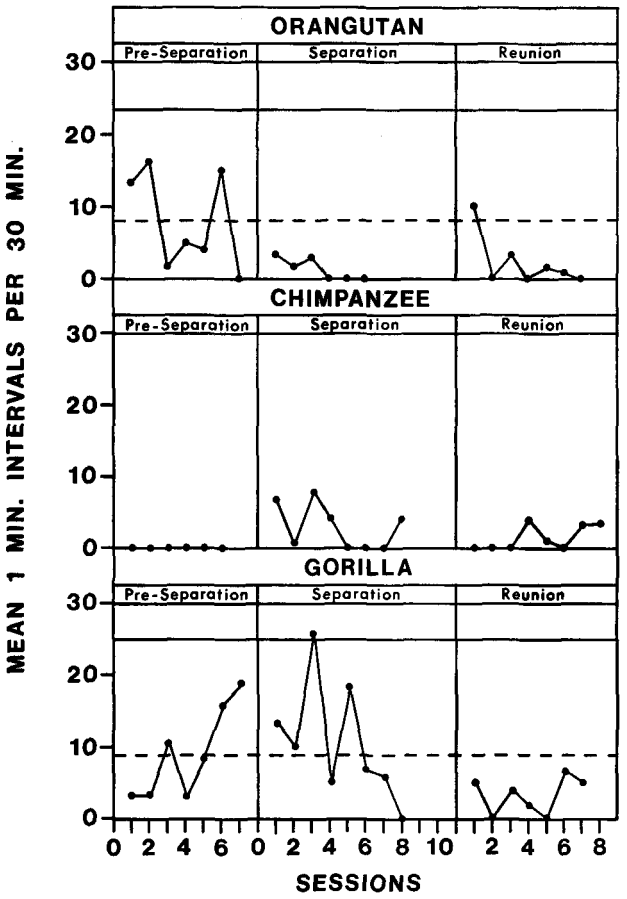


Fig. 3. Levels of positive interactive behaviors for orang-utan, chimpanzee and gorilla infants and their cagemates prior to separation, during separation and following reunion. Broken and solid horizontal lines are defined as in Figure 1.

interactive behaviors took place between the orang-utan infant and its juvenile cagemate (Fig. 3). The behaviors took the form of social play (slapping, chasing, climbing) and visual social examination of the cagemate by the infant. The chimpanzee infant did not interact with its adult female cagemate (0.0%). High levels (30.0%) of positive interactive behaviors also took place between the gorilla infant and its female sibling cagemate. These behaviors were in the form of visual social examination, social play (slapping, tumbling) and infant-initiated back-riding.

Separation

During the two-week period of separation, the juvenile male orang-utan made frequent attempts to interact with the infant, climbing in tandem with the infant, pulling it from the bars, examining various parts of its body and rubbing its genitals on the infant. The infant responded to these overtures throughout the first week of separation by climbing away from

the cagemate and vocalizing in distress. Although total contact with the juvenile cagemate increased, positive interactive behaviors fell below the pre-separation mean. In order to prevent negative interactive behaviors, the juvenile male was removed during each observation session during the second week of separation and the infant was observed alone. When observed immediately prior to reunion, however, the infant was clinging ventro-ventrally to the juvenile male (adoption).

When the chimpanzee infant was returned to its home cage immediately following separation, it attempted to initiate contact by walking bipedally with outstretched arms toward its cagemate. After initial avoidance of the distressed infant, the cagemate accepted positive interaction with the infant (adoption). Positive interactive behaviors between the chimpanzee infant and its cagemate increased significantly (S.D. = 0.0, $df = 5$, $p<.05$) for sessions 1–4 and session 8 of separation.

When the gorilla infant was returned to its home cage, it immediately initiated and was accepted into contact with its cagemate (adoption). The infant assumed an active role by initiating and breaking contact. Although session 3 was the only session with a significantly higher level of positive interactive behaviors (S.D. = 6.6, $df = 6$, $p<.05$), levels of positive interactive behaviors were higher during the first week of separation than during the second week of separation.

Infant Behaviors

Agitation

Changes in levels of agitation during the period of separation compared to levels prior to separation give some indication of the effects of the separation experience on the infant related to protest. Agitation in the infant was defined by distress vocalizations and active locomotor behaviors. Active locomotor behaviors were defined as locomotion not associated with play or object examination (i.e., active climbing for the orang-utan, active climbing and walking for the chimpanzee and gorilla). The level of agitation was calculated as the mean of 1-min intervals of the 30-interval (min) test when either distress vocalizations or active locomotor behaviors occurred.

Pre-separation

During the two weeks prior to separation, the infants spent a considerable amount of time climbing independently or walking in association with examination of objects in the cage (e.g., the bars, food scraps, etc.). Agitation was low for the orang-utan (9.0%), chimpanzee (1.3%) and gorilla (0.7%) (Fig. 4).

Separation

Agitation increased significantly to maximum values during session 1 of separation for the orang-utan (S.D. = 1.6, $df = 6$, $p<.01$), chimpanzee (S.D. = 0.4, $df = 5$, $p<.01$) and gorilla (S.D. = 0.3, $df = 6$, $p<.01$). Levels of agitation displayed by the orang-utan and chimpanzee infants remained significantly higher for sessions 1–4 with a steady decline and

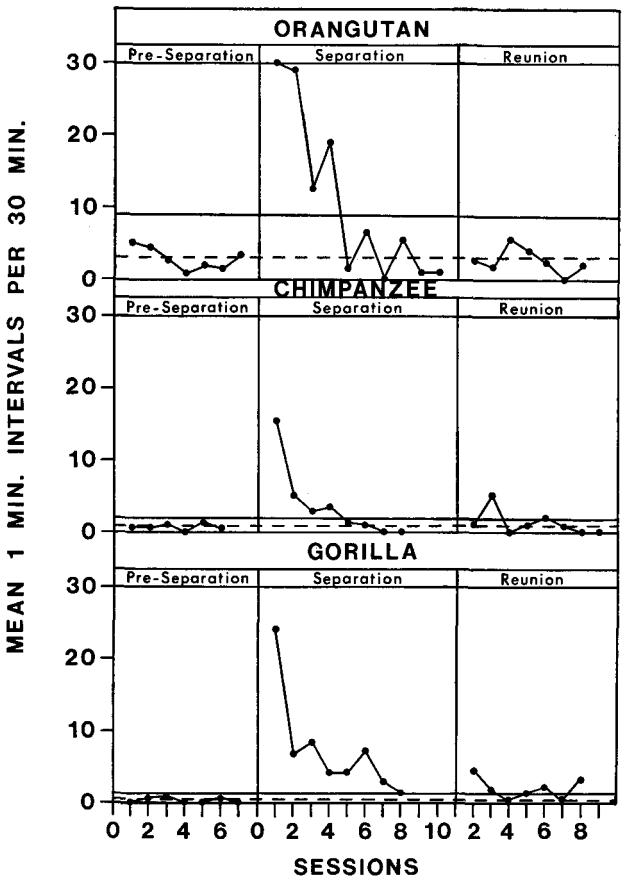


Fig. 4. Levels of agitation for orang-utan, chimpanzee and gorilla infants prior to separation, during separation and following reunion. Broken horizontal lines represent mean levels and solid horizontal lines represent 99% confidence intervals for pre-separation levels.

return to baseline by session 5. For these two infants, agitation remained at baseline for the second week of separation. Although sessions 1–8 were significantly higher for agitation in the gorilla infant, a steady decline in agitation occurred after session 1.

Reunion

For the entire two-week period of reunion, agitation continued at relatively low levels for all three species. Although agitation was significantly high for session 2 in the chimpanzee and for sessions 1–2, 5 and 7 in the gorilla, the levels were considerably lower than agitation during session 1 of separation. These sessions were, in part, significant as a function of the near-zero level observed during pre-separation.

Behavioral Depression

Changes in the levels of behavioral depression during the period of separation compared to

levels prior to separation give some indication of the effects of the separation experience on the infant related to despair. Behavioral depression in the infant was defined by a lack of locomotion (i.e., inactive hanging for the orang-utan, inactive sitting and reclining for the chimpanzee and gorilla). Inactive behaviors were defined as behaviors not associated with ventro-ventral contact, play or object examination. The level of behavioral depression was calculated as the mean of 1-min intervals of the 30-interval (min) test when inactive behaviors occurred.

Pre-separation

As noted above, during the two weeks prior to separation, the infants spent a considerable amount of time climbing independently or walking in association with the examination of objects and with play. Behavioral depression was relatively low for the orang-utan (2.0%), chimpanzee (7.3%) and gorilla (1.7%) (Fig.5).

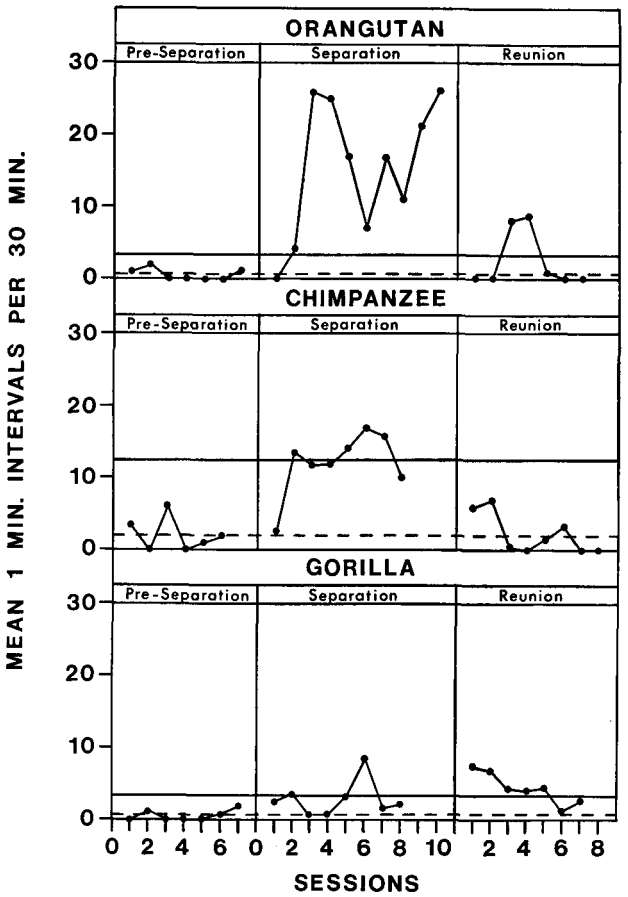


Fig. 5. Levels of behavioral depression for orang-utan, chimpanzee and gorilla infants prior to separation, during separation and following reunion. Broken and solid horizontal lines are defined as in Figure 4.

Separation

In each species, behavioral depression was significantly higher at the beginning of session 2, although minimally higher for the gorilla (S.D. = 0.8, $df = 6, p < .01$). Levels of behavioral depression for the orang-utan increased significantly to a maximum (S.D. = 0.8, $df = 6$, $p < .01$) during session 3 (day 3) of separation. Levels of behavioral depression fluctuated thereafter throughout the period of separation, with high levels again noted during the last few sessions. The chimpanzee infant showed significant (S.D. = 2.5, $df = 5, p < .01$) levels of behavioral depression for session 2 and sessions 5–7, with an increase during the second week of separation after a return to near baseline levels during sessions 3 and 4. The gorilla infant, however, exhibited the highest level of depression (S.D. = 0.8, $df = 6, p < .01$) on session 6 (day 8).

Reunion

During the entire two-week period of reunion for the orang-utan and chimpanzee, levels of behavioral depression remained relatively low compared to the peaks during separation. Behavioral depression was elevated for sessions 3 and 4 for the orang-utan, returned to baseline levels for the chimpanzee, and remained above baseline levels for several sessions for the gorilla. The fact that significant levels of behavioral depression were noted during reunion in the orang-utan and gorilla is, in part, a function of the near-zero levels observed during pre-separation in these species.

DISCUSSION

The present findings illustrate a commonality among great ape infants regarding changes in behavior in response to maternal separation and reunion. Agitation, consisting of distress vocalizations and active locomotor behaviors, which was clearly observed in all three infants immediately following separation, is comparable to the behavioral phenomenon described as "protest" in other nonhuman primate and human infants (BOWLBY, 1973; MINEKA & SUOMI, 1978; REITE, 1977; ROSENBLUM & PLIMPTON, in press; SPITZ, 1946). Persistence in significant levels of agitation beyond the first few days was apparent, moreover, in all three infants. This differs from the typically short-lived responses of monkeys, and is similar to those of human children and other apes that have been studied (BARD & NADLER, in press).

The initial period of agitation in response to maternal separation was followed by a period of increased behavioral depression. Behavioral depression, consisting of inactive behaviors, is similar to the behavioral reaction described as "despair" in separated infants of other species (BOWLBY, 1973; MINEKA & SUOMI, 1978; REITE, 1977; ROSENBLUM & PLIMPTON, in press; SPITZ, 1946).

The reason for the less intense behavioral depression of the gorilla infant in response to separation from its mother is not clear due to a number of confounding variables (i.e., age, sibling cagemate, species differences and individual differences). Consideration of the relationship formed between the infant and its sibling cagemate during separation, however, is consistent with the suggestion that this close relationship could have attenuated the depressed behavioral response to separation from its mother (Fig. 3).

It is also noteworthy that both the chimpanzee and gorilla infants, which displayed the less intense behavioral depression in response to maternal separation, had adult females as cagemates, whereas the orang-utan infant had a juvenile male as a cagemate. Perhaps adult females function as better surrogate mothers to infants than do young males during periods of distress. These data suggest that a positive, compatible relationship with another animal during the period of separation may lessen the behavioral depression reaction to separation from the infant's mother. In order to obtain a more definitive statement regarding the effect of a companion during maternal separation, an approach which compares the behavioral responses of separated infants, both with and without cagemates, would be required.

A clear and common feature of reunion for all three infants was an initial period during which they displayed temporary avoidance of their mothers upon reunion. The great ape infants, therefore, showed a behavioral reaction, albeit of short duration, that is similar to the response displayed by human children following maternal separation and reunion. The formation of new social relationships and the initial avoidance of the mother upon reunion constitute the main characteristics of "detachment" described in human children (BOWLBY, 1973). The transfer of attachment to another conspecific during separation also has been reported in several species of monkeys (DOLHINOW, 1980; JAY, 1965; ROSENBLUM, 1971; SPENCER-BOOTH, 1964; ZUCKERMAN, 1932).

In certain respects, the results of this study more closely resemble the results of comparable studies with human infants than those with monkeys. Whereas many of the nonhuman primates studied exhibited behavior characterized as agitation and behavioral depression following separation, the infant monkeys typically did not exhibit detachment in a free-access situation, in which the reestablishment of attachment between the mother and infant occurs immediately following reunion (MINEKA & SUOMI, 1978). It was suggested that the rapid retrieval of the infant by the mother immediately following reunion masks the behavioral response displayed by the infant (NADLER & CODNER, 1983). Research with pigtail macaques, in which mothers were placed in wire-mesh restraint boxes which prevented them from making contact with their infants during reunion, revealed that some of these infants showed signs of temporary depression in the presence of their mothers and were socially withdrawn (PLIMPTON & ROSENBLUM, in press; ROSENBLUM, 1978). These and related data (ABRAMS, 1969; ROSENBLUM & ALPERT, 1977) led to the proposal that the environment in which separation and reunion take place appears to be an important determinant of whether or not an infant displays a detachment response (ROSENBLUM & PLIMPTON, in press). The responses described as detachment in the monkeys (PLIMPTON & ROSENBLUM, in press), gorillas (HOFF et al., 1979; NADLER & GREEN, 1975) and the three great ape infants of the present study seem quite analogous to those reported for human infants.

Although there was a relatively brief initial period when the infants appeared to reject the returned mothers, an intensification of the mother-infant relationship followed reunion in all three. The subsequent intensification of attachment between the mother and infant following reunion is typical of both human and nonhuman primates. This heightened attachment was characterized by significant increases in the levels of ventro-ventral and nipple contact. The older age of the gorilla infant may have accounted for less intensified attachment and a faster recovery from the separation experience.

The strongyloides hyperinfection that resulted in the death of the orang-utan infant after the study was completed raises the question of whether this condition influenced the

infant's behavior during reunion and its health afterwards. Since the possibility of an influence by the treatment cannot be ruled out completely, it would be prudent to maintain close surveillance of the health of infants in any further research of this type.

Although there is some degree of variability among studies investigating the effects of maternal separation and reunion on infants (LEWIS et al., 1976), there is widespread occurrence among all primates studied of behavioral reactions consisting of an initial agitation phase, followed by a behavioral depression phase during separation and an intensification of attachment following reunion. A detachment phase, however, is found more commonly in humans and in the great apes. This finding and that of intermittent agitation during separation support the notion that the close biological relationship between apes and humans makes apes especially useful for investigating certain behavioral phenomena of interest in humans.

Acknowledgements. This research was supported by National Science Foundation Grant No. BNS 79-23015 to R.D.N. and U.S. Public Health Service Grant No. RR-00165 (Division of Research Resources, National Institutes of Health) to the Yerkes Regional Primate Research Center of Emory University. Support was also provided by Emory University College Work Study Program funded by the Department of Education, HEW. The authors thank Ms. K. A. BARD for assistance with statistical analyses, Ms. P. PLANT for generous typing of the manuscript, Mr. J. R. ROBERTS for facilitating the behavioral observations, and Drs. D. A. EDWARDS, D. R. STOKES and K. WALLEN for comments on an earlier version of this paper.

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—Received *June 28, 1983*; Accepted *October 27, 1983*

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